

AKHILESH KUMAR MISHRA

PYTHON FULL STACK DEVELOPER | DJANGO · FASTAPI · REST APIS

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India
12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic web services into asynchronous **FastAPI** and **Django** microservices, improving reliability for **2 million+ users**.
- Scaled transaction orchestration services for AEPS, MATM, and VATM platforms on **Django REST Framework**, sustaining **500,000+ daily transactions**.
- Secured APIs with **JWT** authentication and role-based access control (**RBAC**) plus OTP verification, halving authentication latency.
- Built an asynchronous task pipeline with **Celery** and **Redis**, clearing **35%** of processing bottlenecks under peak traffic.
- Tuned **PostgreSQL** and **MySQL** schemas with indexing and query partitioning, lowering transaction query latency by **45%**.
- Streamed live transaction status through **FastAPI WebSockets**, delivering updates to monitoring clients in under **500ms**.

Freelance Software Engineer

Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform
06/2023 – 11/2023 | Remote, India

- Developed a real-time workforce telemetry and geofencing system on **FastAPI** and **SQLAlchemy** (asyncio), serving **1,000+ active field agents**.
- Built a high-throughput asynchronous GPS ingestion API, sustaining **10,000+ concurrent location pings at 99.9% uptime**.
- Delivered the live telemetry dashboard in **React.js** against FastAPI WebSocket endpoints, cutting UI load times by **30%**.

SELECTED PROJECTS

Facial Recognition Attendance System

Automated attendance tracking in Python with OpenCV (Haar Cascade and LBPH) face recognition, MySQL persistence, and Pandas-generated Excel reports.

STACK: Python / OpenCV / Pandas / Tkinter / MySQL

Himalayan Edges Commerce Platform

PWA-enabled e-commerce platform on Django and Django REST Framework, deployed to AWS, lifting page speed and client usage by 30%.

STACK: Python / Django / Django REST Framework / SQLite / AWS / PWA

Async Telemetry Ingestion & Reporting Service

FastAPI ingestion service with Celery and Redis task queues writing to partitioned PostgreSQL tables, plus Pandas reporting jobs over aggregated telemetry.

STACK: Python / FastAPI / Celery / Redis / PostgreSQL / Pandas

Persistent Task Management Dashboard

Python desktop utility for task planning, category filtering, and JSON persistence, structured on a clean MVC separation.

STACK: Python / Tkinter / JSON / MVC

SUMMARY

Software Development Engineer specializing in Python Full Stack development, building secure, asynchronous backend architectures for high-volume FinTech platforms. Deep expertise in Python, Django, Django REST Framework, FastAPI, Celery, SQLAlchemy, and PostgreSQL/MySQL schema design. Migrated monoliths to async microservices serving 2 million+ users, built Celery and Redis pipelines that cleared 35% of peak-load processing bottlenecks, and streamed live transaction status over FastAPI WebSockets in under 500ms.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 – 2023 | Kanpur, UP

SKILLS

CORE PYTHON

Python Django Django REST Framework FastAPI
Celery SQLAlchemy asyncio

APIS & SECURITY

RESTful APIs JWT authentication OAuth RBAC
FastAPI WebSockets API versioning

DATA & MESSAGING

PostgreSQL MySQL SQLite Redis Celery queues
indexing partitioning query optimization

TOOLING & FRONTEND

Git GitHub Docker Linux Postman CI/CD
React.js JavaScript (ES6+) HTML5 CSS3

CERTIFICATIONS

Database Systems & SQL Certification
Data Structures & Algorithms Certification
IIT Kanpur Cyber Security Certification
Advanced MERN Stack Developer Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.